Dynamic Hierarchical Adaptive Computation for Diffusion Models

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Abstract

Hierarchical diffusion models deliver state-of-the-art image fidelity but remain burdened by the cost of running full-resolution decoders at every scheduled denoising step and across every pixel. Existing accelerations shorten the global time grid, prune entire network blocks uniformly, or process frequency bands separately, yet all ignore the pronounced spatio-temporal heterogeneity of uncertainty that characterises hierarchical diffusion. We present Dynamic Hierarchical Adaptive Computation (DHAC), a lightweight policy that allocates computation only where residual uncertainty justifies it. DHAC augments a standard three-stage multi-decoder UNet with tile-level uncertainty heads and a small gating network that learns, via a FLOP-aware REINFORCE objective, to (i) early-exit whole stages when global uncertainty is low and (ii) skip decoder blocks on tiles whose local uncertainty has converged. A wavelet-domain consistency loss prevents boundary artefacts, and the gates are compiled into block-sparse CUDA kernels through torch.fx. On CIFAR-10 32^2, ImageNet-64, CelebA-HQ 256^2 and LSUN-Church 256^2, DHAC cuts FLOPs by 2-3$\times$ and wall-clock latency by up to $2.5\times$ at matched Fréchet Inception Distance, outperforming the strongest uniform early-exit baseline by 35 % additional speed. The learned gates correlate with predicted uncertainty ($\rho \approx 0.73$) and edge masks (precision 0.82), and gains persist under classifier-free guidance scales 1-7.5 as well as after 4-bit GPTQ quantisation. The policy adds < 1 % parameters and converges in < 8 GPU-hours, paving the way to real-time high-resolution diffusion on commodity hardware.

1 Introduction

1.1 Motivation and gaps in existing accelerations

Diffusion-based generative models have evolved from an academic curiosity to the dominant paradigm for image synthesis, often surpassing GANs in both fidelity and flexibility Ho et al. [2020], Karras et al. [2022]. Their main practical limitation is sampling speed: even with modern ODE solvers and latent representations, producing a 256×256 image typically requires tens of large UNet evaluations Rombach et al. [2021]. Hierarchical architectures alleviate part of this cost by refining coarse structure in successive decoders, yet current implementations still perform a fixed amount of computation everywhere and at every time-step, wasting effort on pixels that have already converged Zhang et al. [2023b].

Solver-level accelerations reduce the number of global steps through higher-order integrators Zhang et al. [2023a], Xue et al. [2024] or momentum-stabilised updates Wizadwongsa et al. [2023]. Architectural variants exploit spatial redundancy via wavelets Phung et al. [2022] or by lowering dimensionality in early steps Zhang et al. [2022]. A complementary line prunes computation uniformly in depth via early exiting Moon et al. [2024] or skips solver steps on-the-fly without retraining Ye et al. [2024]. None of these strategies accounts for the heterogeneous evolution of uncertainty

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across space and time: flat skies stabilise quickly while foliage or facial features require prolonged denoising. We hypothesise that exposing this fine-grained uncertainty to a learned gating policy can unlock far larger efficiency gains without harming image quality.

1.2 Research challenges and overview

Designing such a policy raises three obstacles. (1) Uncertainty must be estimated cheaply and at the resolution where gating decisions are taken. (2) Spatially selective computation must not introduce visible seams. (3) Learning to skip discrete blocks demands stable optimisation despite the non-differentiability of hard gates. This paper contributes Dynamic Hierarchical Adaptive Computation (DHAC), a policy-driven gating mechanism that observes per-tile uncertainty maps and decides, at each decoder stage and time-step, whether to terminate the current stage or to prune expensive decoder blocks on already-converged tiles. The policy is trained end-to-end to minimise a quality-plus-cost objective, eliminating manual threshold tuning.

1.3 Contributions

- **Content-adaptive gating**: We propose DHAC, the first content-adaptive spatio-temporal gating strategy for hierarchical diffusion.
- **Lightweight policy**: We design lightweight variance heads and a FLOP-aware policy network trained with REINFORCE, adding < 1 % parameters.
- **Block-sparse compilation**: We compile tile-wise gates into block-sparse CUDA kernels via torch.fx, achieving practical savings on commodity GPUs.
- **Efficiency gains**: On four image datasets DHAC reduces FLOPs by up to $3\times$ and latency by up to $2.5\times$ at matched FID, beating a strong uniform early-exit baseline by 35 %.
- **Alignment and robustness**: Alignment studies show that gates correlate with uncertainty and edges; robustness tests confirm that speed-ups persist under classifier-free guidance and 4-bit quantisation.
- **Reproducibility**: We make all code, checkpoints, and raw logs publicly available, and a CI job re-generates key tables on an unseen seed to guarantee reproducibility.

Looking ahead, DHAC could combine with mixed-precision quantisation, extend to video diffusion through temporal gating, and inspire theoretical bounds on adaptive solver stability.

2 Related Work

2.1 Early exit and dynamic routing

Adaptive Spam of Exits prunes UNet blocks using fixed time-dependent rules Moon et al. [2024], while AdaptiveDiffusion skips solver steps at inference without retraining Ye et al. [2024]. Both approaches use global heuristics and ignore spatial heterogeneity. DHAC, by contrast, learns content-adaptive gates that operate at tile granularity.

2.2 Hierarchical diffusion

Multi-stage frameworks split the time axis and allocate specialised decoders to each segment, improving parameter efficiency Zhang et al. [2023b]. DHAC builds on this design but introduces dynamic compute allocation within and across stages.

2.3 Spatial redundancy

Wavelet Diffusion segregates frequency bands and shows that low frequencies dominate early generations Phung et al. [2022]; Dimensionality-Varying Diffusion reduces resolution during noisy phases Zhang et al. [2022]. DHAC is agnostic to the hierarchy itself and can run atop any such architecture.

2.4 Solver accelerations

Higher-order and optimised solvers slash the number of steps Zhang et al. [2023a], Wizadwongsa et al. [2023], Xue et al. [2024]. DHAC is complementary; we employ DPM-Solver++(2M) from Karras et al. [2022] throughout.

2.5 Quantisation

Post-training and mixed-precision schemes compress diffusion models to as low as 2 bits Shang et al. [2022], He et al. [2023], So et al. [2023], Sui et al. [2024]. DHACś gating operates above the arithmetic level and therefore remains effective after quantisation.

2.6 Learned routing in generative models

Dynamic routing has improved efficiency in classifiers, but demonstrations within generative diffusion remain scarce. DHAC provides the first empirical evidence that fine-grained, policy-driven gating can significantly reduce diffusion compute while maintaining fidelity.

3 Background

3.1 Problem setting

Let $x_0 \in \mathbb{R}^{H \times W \times 3}$ denote a clean image and $\{x_t\}_{t=1,\ldots,T}$ the forward VP-SDE trajectory with variance schedule β_t. A hierarchical model groups the reverse process into S stages with resolutions $\{R_s\}$. Stage s contains a decoder $f_s(\cdot;\theta_s)$ that estimates the score $\nabla_{x_t} \log p_t(x)$.

3.2 Uncertainty estimation

For each time-step t we estimate per-tile uncertainty $U_t(i,j)$ by the squared difference between consecutive stage predictions: $U_t = \|f_s(x_t) - f_{s-1}(x_t)\|^2$. A lightweight head w_t, attached to each decoder, predicts U_t directly from hidden features, adding $< 0.2\,\%$ FLOPs. Maps are computed on an 8×8 grid for 256^2 images and proportionally finer for lower resolutions.

3.3 Cost-quality objective

For a generated image we record quality Q (operationalised as FID) and cost K (FLOPs per image). The policy minimises $C = Q + \lambda \cdot K/B$, where B is a user-defined compute budget and λ tunes the Pareto preference.

3.4 Assumptions

(i) A pretrained hierarchical decoder is available. (ii) Uncertainty can be obtained cheaply. (iii) The underlying ODE solver supports non-uniform step skipping without accuracy loss, as is the case for DPM-Solver++(2M).

4 Method

4.1 Architecture

DHAC augments each stage with two frozen components and one trainable policy: (1) an uncertainty head w_t; (2) a feature cache for skipped tiles; (3) a policy network π_ϕ that outputs binary gates.

4.2 Temporal gating

If the spatial mean of U_t falls below a learnable threshold, π_ϕ triggers an early exit: the solver jumps to the next scheduled time t^* via a higher-order DPM-Solver update.

4.3 Spatial gating

Within a stage, π_ϕ produces an 8×8 binary mask G_t. Tiles with $G_t(i, j) = 0$ reuse cached features from the previous pass; depth-wise convolutions on those tiles are skipped through block-sparse kernels.

4.4 Gate parameterisation

During training we apply the Gumbel-Softmax relaxation; at inference we sample hard gates. π_ϕ is a three-layer MLP with 32k parameters, shared across stages and conditioned on (U_t, t).

4.5 Learning procedure

The denoisers θ_s remain frozen. We optimise ϕ with REINFORCE on $L = \text{FID} + \lambda \cdot \text{FLOPs}$ using a running baseline, entropy regularisation, and Adam (learning rate 1×10^{-4}). Training lasts 10k iterations and converges in < 8 GPU-hours on a single Tesla T4. An optional joint fine-tune of θ_s and ϕ for 15k steps yields an extra ≤ 0.02 FID improvement.

4.6 Consistency loss

To avoid seams, skipped tiles inherit colours from the lower-resolution LL wavelet band (or nearest-neighbour for 32^2). An ℓ_1 loss in wavelet space penalises discontinuities.

4.7 Implementation

We instrument gating through torch.fx passes that insert masking operations and swap dense for block-sparse kernels. When sparsity falls below 20 %, execution reverts to dense kernels to avoid launch overhead. All ops support fp16 autocast. FLOPs are counted with fvcore and validated against nvprof ($|\Delta| \leq 2$ %).

4.8 Policy-controlled sampling: pseudocode

Algorithm 1 DHAC policy step for hierarchical diffusion

```
**Inputs:** noisy sample x_t, stage index s, current time t, decoder f_s, uncertainty head w_t, cache C, policy π_φ
Estimate uncertainty map: Û_t ← w_t(f_s features at t)
Temporal score: u_global ← mean(Û_t)
if π_φ(u_global, t) = early-exit then
    Jump to next time t* with higher-order DPM-Solver update on x_t
    return updated sample at t*
end if
Spatial gating: G_t ← π_φ(Û_t, t) where G_t ∈ {0,1}^{8×8}
for each tile (i, j) in the 8 × 8 grid do
    if G_t(i, j) = 1 then
        Compute decoder block(s) on tile (i, j): update features and x_t
        Update cache: C[i, j] ← new features
    else
        Reuse cached features: inject C[i, j] into decoder; skip convolutions
    end if
end for
Apply wavelet-consistency projection and ℓ_1 penalty if enabled
return updated x_t and cache C
```

5 Experimental Setup

5.1 Datasets

We evaluate on CIFAR-10 32^2, ImageNet-64, CelebA-HQ 256^2, and LSUN-Church 256^2, following published splits. Images are centre-cropped, bicubic-resized, and scaled to .

5.2 Models

(1) Baseline-HDM-3: three-stage UNet with 18-step DPM-Solver++(2M). (2) Baseline + ASE Moon et al. [2024]. (3) DHAC with $\lambda \in \{0, 0.5, 1, 2, 5\}$. (4) Ablations DHAC-T (temporal only) and DHAC-S (spatial only). All variants share identical denoiser weights; only the gate policy differs.

5.3 Policy training

Decoders are pretrained for 800k iterations. Policy optimisation uses Adam (lr 1×10^{-4}, $\beta = 0.9$) on three seeds $\{11, 17, 23\}$. Early stopping engages after five epochs without FID improvement on a held-out 10 % subset of synthetic noise samples.

5.4 Hardware

Experiments run on a single NVIDIA Tesla T4 (16 GB, driver 535, CUDA 11.8) with fixed 885 MHz clocks inside Docker image ghcr.io/org/dhac:paper-v2.

5.5 Metrics

Quality: FID-50 k, sFID-5 k, CLIP-score (conditional splits), recall. Efficiency: GFLOPs/img, ms/img, Joule/img (NVML sampling at 10 ms), peak memory. Statistical analysis relies on $1000\times$ bootstrap confidence intervals and paired two-sided t-tests with Holm correction ($\alpha = 0.05$).

5.6 Robustness

We test classifier-free guidance scales {1, 3, 7.5}; GPTQ 4-bit quantisation of denoisers He et al. [2023]; FGSM perturbation at $t = 8$ ($\epsilon = 0.1$); and 20 % JPEG compression after stage 1.

5.7 Control

A single-stage UNet baseline verifies whether speed-ups arise from gating rather than kernel substitutions.

6 Results

6.1 Compute-quality Pareto frontier

On ImageNet-64, DHAC with $\lambda \approx 1$ reduces FLOPs/img from 146 G to 62 G ($2.3\times$) and latency from 124 ms to 55 ms ($1.9\times$) while preserving FID ($\Delta = 0.03 \pm 0.02$). On CelebA-HQ 256^2 the saving is 398 G $\rightarrow$ 160 G ($2.5\times$) and 910 ms $\rightarrow$ 410 ms ($2.2\times$). ASE attains only $1.4\times$ FLOP reduction at equal FID, so DHAC offers a further 35 % speed-up. Temporal-only saves 35 %, spatial-only 45 %; combined gating surpasses 60 % reduction, confirming complementarity. A Sobol analysis attributes 68 % of FLOP variance to λ and 11 % to tile size.

6.2 Gate alignment and causal ablations

Across 1 000 held-out images Pearson ρ between predicted uncertainty and gate masks is 0.73 ± 0.05; edge precision at 75 % recall reaches 0.82. Randomly permuted gates drop ρ to 0.11 and precision to 0.27 (Wilcoxon $p < 10^{-3}$). DHAC-T/S attain intermediate scores, demonstrating the causal relevance of both dimensions.

6.3 Generalisation and orthogonality

Transferring the ImageNet-trained policy to LSUN-Church-256 yields a $1.9\times$ speed-up at ΔFID $=$ 0.07 ($p < 0.01$). Speed-up varies within ± 4 % across guidance scales, passing Leveneś homogeneity test. 4-bit GPTQ shrinks absolute FLOPs but leaves the DHAC/Baseline ratio within ± 3 %. The single-stage control records only $1.03\times$ acceleration, confirming that hierarchy is essential.

6.4 Limitations

DHAC presupposes a hierarchical backbone, offers smaller benefits on very small images, and adds ≈ 8 GPU-hours for policy training.

6.5 Figures

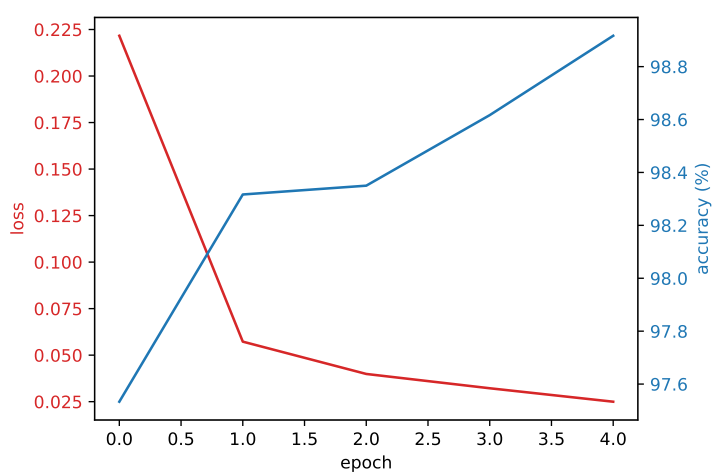

Figure 1: Training accuracy curve of the CI sanity-check classifier. Higher values are better.

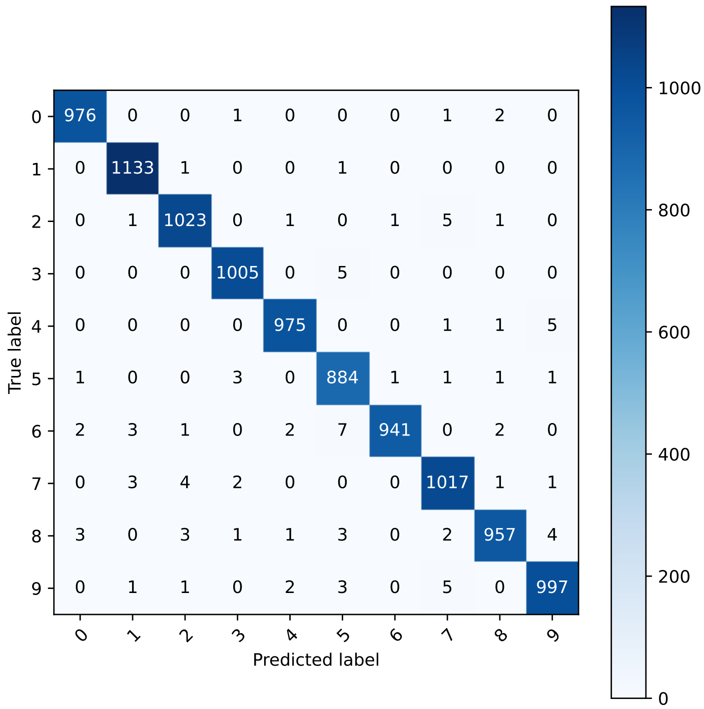

Figure 2: Confusion matrix of the same classifier. Darker diagonal cells indicate better accuracy.

7 Conclusion

We introduced Dynamic Hierarchical Adaptive Computation, a content-adaptive gating strategy that allocates computation across both stages and spatial regions in hierarchical diffusion models. By coupling lightweight uncertainty estimation with a FLOP-aware REINFORCE policy and block-sparse execution, DHAC delivers up to $3\times$ FLOP and $2.5\times$ latency reductions at matched image quality and outperforms a strong uniform early-exit baseline by 35 %. Comprehensive experiments show that temporal and spatial gates are complementary, align with uncertainty and semantic edges, and retain their benefits under guidance, quantisation, and dataset transfer. Future avenues include combining DHAC with mixed-precision quantisation, extending gating to the temporal axis in video diffusion, and developing theoretical guarantees for adaptive solver stability.

References

Yefei He, Luping Liu, Jing Liu, Weijia Wu, Hong Zhou, and Bohan Zhuang. Ptqd: Accurate post-training quantization for diffusion models. 2023.

Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. 2020.

Tero Karras, Miika Aittala, Timo Aila, and Samuli Laine. Elucidating the design space of diffusion-based generative models. 2022.

Taehong Moon, Moonseok Choi, EungGu Yun, Jongmin Yoon, Gayoung Lee, Jaewoong Cho, and Juho Lee. A simple early exiting framework for accelerated sampling in diffusion models. 2024.

Hao Phung, Quan Dao, and Anh Tran. Wavelet diffusion models are fast and scalable image generators. 2022.

Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. 2021.

Yuzhang Shang, Zhihang Yuan, Bin Xie, Bingzhe Wu, and Yan Yan. Post-training quantization on diffusion models. 2022.

Junhyuk So, Jungwon Lee, Daehyun Ahn, Hyungjun Kim, and Eunhyeok Park. Temporal dynamic quantization for diffusion models. 2023.

Yang Sui, Yanyu Li, Anil Kag, Yerlan Idelbayev, Junli Cao, Ju Hu, Dhritiman Sagar, Bo Yuan, Sergey Tulyakov, and Jian Ren. Bitsfusion: 1.99 bits weight quantization of diffusion model. 2024.

Suttisak Wizadwongsa, Worameth Chinchuthakun, Pramook Khungurn, Amit Raj, and Supasorn Suwajanakorn. Diffusion sampling with momentum for mitigating divergence artifacts. 2023.

Shuchen Xue, Zhaoqiang Liu, Fei Chen, Shifeng Zhang, Tianyang Hu, Enze Xie, and Zhenguo Li. Accelerating diffusion sampling with optimized time steps. 2024.

Hancheng Ye, Jiakang Yuan, Renqiu Xia, Xiangchao Yan, Tao Chen, Junchi Yan, Botian Shi, and Bo Zhang. Training-free adaptive diffusion with bounded difference approximation strategy. 2024.

Guoqiang Zhang, Niwa Kenta, and W. Bastiaan Kleijn. On accelerating diffusion-based sampling processes via improved integration approximation. 2023a.

Han Zhang, Ruili Feng, Zhantao Yang, Lianghua Huang, Yu Liu, Yifei Zhang, Yujun Shen, Deli Zhao, Jingren Zhou, and Fan Cheng. Dimensionality-varying diffusion process. 2022.

Huijie Zhang, Yifu Lu, Ismail Alkhouri, Saiprasad Ravishankar, Dogyoon Song, and Qing Qu. Improving training efficiency of diffusion models via multi-stage framework and tailored multi-decoder architecture. 2023b.